

Maximum Nullity of Cyclic Covers of Graphs

A monodromy ceiling, and the zero forcing
number of $P(n, k)$

Research Binder

Aryan Padarthy · Mathematics

Compiled 27 September 2026

Contents

- I. Official Abstract
- II. Research Plan
- III. Research Paper
- IV. Research Log, dated, including every dead end
- V. Verification Output, 69 checks
- VI. A Certification Transcript

Every claim in the paper is labelled by the standard it meets: proved, proved by an exact certificate, proved by interval certification, or numerical. Claims that have been refuted or withdrawn, including the author's own, are listed explicitly in the paper's Status section and dated in the log.

I. Official Abstract

Zero forcing is a colouring process on a graph: a coloured vertex with exactly one uncoloured neighbour forces that neighbour to colour. The zero forcing number $Z(G)$ is the size of the smallest starting set that eventually colours everything, and it bounds the maximum nullity $M(G)$ over symmetric matrices whose off-diagonal pattern is the graph. For the generalized Petersen graphs $P(n,k)$, a 2020 paper claimed $Z(P(n,3))=8$ for all n at least 12. A July 2026 correction note showed this is false, and recorded two open problems: the stabilization threshold, and any rigorous lower bound for k at least 4.

I prove that eliminating the inner coordinates of any matrix carrying the $P(n,k)$ pattern leaves a single linear recurrence of order exactly $2k+2$. Its nullity is therefore at most $2k+2$ the same number as the best known upper bound on Z . The matrix method thus either settles Z completely or cannot settle it at all.

Reaching that ceiling with a rotation-invariant matrix provably confines n to one divisibility class, containing at most one prime. Assembling the matrix instead from tiles that each contribute the identity to the monodromy removes the restriction: tiles of every length in $[L,2L)$ settle every n at least L . I certified 55 such tiles by interval arithmetic.

Consequently $Z(P(n,2))$ and $Z(P(n,3))$ are now determined for every n , the latter for the first time; $Z(P(n,3))=8$ exactly for n at least 13, settling the threshold; and $Z(P(n,4))=10$ for every n at least 30.

II. Research Plan

Project. Maximum Nullity of Cyclic Covers of Graphs: A Monodromy Ceiling and the Zero Forcing Number of Generalized Petersen Graphs **Category.** Mathematics **Finalist.** Aryan Padarthy

A. Rationale

Zero forcing began in two unrelated places: controllability of quantum spin systems, and placement of phasor measurement units in electrical grids. It was then connected to linear algebra by the inequality $M(G) \geq Z(G)$, where $M(G)$ is the largest nullity attainable by a symmetric matrix whose off-diagonal support is the edge set of G . The inequality makes Z a combinatorial ceiling on a purely algebraic quantity, and makes lower bounds on Z obtainable by exhibiting a matrix.

The generalized Petersen graphs $P(n,k)$ are a standard cubic test family. Rashidi, Shajareh Poursalavati and Tavakkoli (2020) proved $Z(P(n,k)) \geq 2k+2$ and determined $Z(P(n,2))$. Their Theorem 3.6, asserting $Z(P(n,3)) = 8$ for $n \geq 12$, is false: Krishnan's July 2026 correction note (arXiv:2607.19412) exhibits $Z(P(12,3)) = 7$, the error being a case analysis that never excludes 7-vertex sets. That note leaves open the stabilization threshold and any rigorous lower bound for $k \geq 4$.

This project addresses those two open problems.

B. Research question

1. Is there an intrinsic ceiling on what a matrix certificate can prove about $Z(P(n,k))$, and where does it sit relative to the known upper bound?
2. Can that ceiling be attained, and by an explicitly checkable matrix?
3. Can $Z(P(n,k))$ be determined for **all** n rather than for sparse families of n ?

C. Procedures

This is a theoretical mathematics project. There are no human participants, no vertebrate animals, no potentially hazardous biological agents, and no hazardous chemicals, activities or devices. All work is proof and computation performed by the finalist on a personal computer.

C1 Structure. Eliminate the inner coordinates from the kernel equations of an arbitrary matrix carrying the $P(n,k)$ pattern and identify the resulting scalar recurrence, its order, and the leading coefficients. Establish when the nullity equals the recurrence order.

C2 Construction. For rotation-invariant matrices, reduce singularity to a polynomial condition on a discrete grid and construct matrices meeting the ceiling. Determine exactly when such a matrix is admissible.

C3 Obstruction. Determine whether the divisibility conditions arising in C2 are intrinsic, and prove the answer rather than observing it.

C4 General n. Construct matrices from repeated blocks whose contributions compose, so that one finite family of blocks settles an infinite set of n .

C5 Certification. Verify every constructed matrix at the highest standard available: exact integer or rational arithmetic where possible, and otherwise an interval (Krawczyk) contraction proof that a true real solution exists. Record which standard each claim meets.

C6 Independent checking. Compute Z directly by exhaustive search over vertex subsets for all sizes reachable, validate that solver against every published value, and reproduce each theorem numerically before relying on it.

D. Data analysis

Results are theorems and explicit matrices, not measurements, so analysis is verification rather than statistics. Every computational claim is re-derived by a single script that reports pass or fail per check. Numerical searches are always wrapped in a bound they cannot legally exceed (the ceiling of C1), so that an impossible answer is detected rather than believed. Where a search fails, the residual and the computational budget are both recorded, because a failure at a small budget is not evidence of impossibility.

E. Risk and safety

None. No laboratory work, no fieldwork, no subjects of any kind.

F. Bibliography

1. S. Rashidi, N. Shajareh Poursalavati, M. Tavakkoli. Computing the zero forcing number for generalized Petersen graphs. *J. Algebra Combin. Discrete Struct. Appl.* 7(2) (2020) 183193.
2. A. Krishnan. A correction to the zero forcing number of the generalized Petersen graphs $P(n,3)$. arXiv:2607.19412 (2026).
3. AIM Minimum Rank Special Graphs Work Group. Zero forcing sets and the minimum rank of graphs. *Linear Algebra Appl.* 428 (2008) 16281648.
4. J. H. Conway, A. J. Jones. Trigonometric diophantine equations. *Acta Arith.* 30 (1976) 229240.
5. I. Niven. *Irrational Numbers*. Carus Mathematical Monographs 11, MAA, 1956.
6. R. E. Moore, R. B. Kearfott, M. J. Cloud. *Introduction to Interval Analysis*. SIAM, 2009. (Krawczyk existence test.)

III. Research Paper

The full paper follows this page. It states, for every claim, which of four standards it meets: proved; proved by an exact certificate; proved by interval certification; or numerical. Its Status section lists every claim of the author's own that has been refuted or withdrawn.